1. 其二进制形式：$-0.00000\,1101$

浮点数规格化表示：$-(0.1101\,00000)\times 2^{-101}$

浮点机中：$[X]_{原}=1.0101$；$1.1101\,0000$

$[X]_{补}=1.1011$；$1.0010\,0000$

$[X]_{反}=1.1010$；$1.0010\,1111$

2. $[A]_{补}=0.0010010$　$[B]_{补}=1.1001\,0$

(1) $[A+B]_{补}=[A]_{补}+[B]_{补}=1.1011110$

所以 $A+B=-\frac{17}{64}$

(2) $[A]_{补}=0.1001\,00$

$[B]_{补}=1.1101111$

$[-B]_{补}=0.0010001$

$[A-B]_{补}=0.1011101$

所以 $A-B=\frac{93}{128}$

3. (1) 原码一位乘. $[x]_{原} = 0.100111$, $[y]_{原} = 1.10101$

$x^* = 0.100111$ $y^* = 0.10101$

$[x \cdot y]_{原}$ 数值部分

$0.000000 \quad 10101$

$+ 0.100111$

0.100111

$0.010011 \quad 11010 \quad \rightarrow 1$

$+ 0.100111$

$0.111010 \quad 1$

$0.011101 \quad 01101 \quad \rightarrow 1$

$0.001110 \quad 10110 \quad \rightarrow 1$

+ 0.100111

0.1101011 0

0.01101011 0110 → 1

~~0.0011010110111 → 1~~

+ 0.100111

0.1101000110 1

0.0110100011101 → 1

$x_0 \oplus y_0 = 1$

故 [x·y]原 = 1.0110001101

补码一位乘

[x]补 = 0.100111

[y]补 = 1.010101

[-x]补 = 1.011001

00.000000 01010

+ 00.100111

00.100111

00.010011 101010 → 1

00.001001 110101 → 1

+ 00.100111

00.110000 11

00.011000 011010 → 1

00.001100 001101 → 1

+ 00.100111

00.100011 001101

00.011001 100110 → 1

00.001100 110011 → 1

+ 11.011001 + [-x]补 所以 [x·y]补 = 1.100101100111

11.100101100111

(2) $[x]_{原}=0,0100011$ $[y]_{原}=0,1000011$ $x^*=0,0100011$ $y^*=0,1000011$

原码一位乘

```
  0,000 000    100011
+ 0,010011
  0,010011
  0,001001 1   110001 →1
+ 0,010011
  0,011100 1
  0,001110 0 1 110000 →1
  0,000111 0 0 111000 →1
  0,000011 1 0 011100 →1
  0,000001 1 1 001110 →1
  0,000000 1 1 100111 →1
+ 0,010011
  0,010100 1 1 100111
  0,001010 0 1 110011 →1
```

$x_0 \oplus y_0 = 0$

得 $[x \cdot y]_{原} = 0,001010011001$

补码一位乘

$[x]_{补}=00,010011$ $[y]_{补}=00,100011$ $[-x]_{补}=11,101101$

```
  00,000000    010011 0
+ 11,101101
  11.101101
  11.110110 1  010100 1 1 →1
  11.111011 0  1010100 0 1 →1
+ 00,010011
  00.001110 1  01
  00.000111 0  1 001010 0 0 →1
  00.000011 1  0 100101 0 0 →1
  00.000001 1  1 110010 1 0 →1
+ 11.101101
  11.101101 1  0 11001
  11.110110 1  1 011001 0 1 →1
+ 00.010011
  00.001010 0  1 011001
```

得 $[x \cdot y]_{补} = 00.001010011001$

4. $[x]_{原}=1.1011$ $[y]_{原}=0.1101$

$x^*=0.1011$ $y^*=0.1101$ $[-y^*]_{补}=1.0011$

```
  0.1011    0.00000
+ 1.0011
-----------
  1.1110    0
  1.1100    0      ←1
+ 0.1101
-----------
  0.1001    01
  0.1110    01     ←1
+ 1.0011
-----------
  0.0001    011
  0.0010    011    ←1
+ 1.0011
-----------
  1.0101    0110
  0.1010    0110   ←1
+ 1.0011
-----------
  1.1101    01100
  1.1010    01100  ←1
+ 0.1101
-----------
  1.1000    011000
```

$x_0 \oplus y_0 = 1$ 所以 $[\frac{x}{y}]_{原}=1.11000$

5. (1) 令 $x=[2^5\times\frac{11}{16}]=2^{101}\times(0.1011)$

$y=[2^4\times(-\frac{9}{16})]=2^{100}\times(-0.1001)$

所以 $[x]_{补}=00,101;\ 00.1011$

$[y]_{补}=00,100;\ 11.0111$

对阶：$[\Delta j]_{补}=[j_x]_{补}-[j_y]_{补}=00,001$

5. (1) 令 $x=[2^5\times\frac{11}{16}]=2^{101}\times 0.101100$

令 $y=[2^4\times(-\frac{9}{16})=2^{100}\times(-0.100100)$

则 $[x]_{补}=00,101;00,101100$.

$[y]_{补}=00,100;11,011100$

① 对阶. $[\Delta x-\Delta y]_{补}=00,001$

则 $[y]_{补}=00,101;11,101110$

② 尾数求和. $[S_x+S_y]_{补}$ 为

$$\begin{array}{r} 00,101100 \\ +\ 11,101110 \\ \hline 00.011010 \end{array}$$

即 $[x+y]_{补}$ 为 $00,101;00,011010$.

③ 规格化. 左规得

$[x+y]_{补}=$ ~~$00,100;$~~ $00,100;00,110100$

(2) 令 $x=[2^{-3}\times\frac{13}{16}]=2^{-011}\times 0.110100$.

令 $y=[2^{-4}\times(-\frac{5}{8})]=2^{-100}\times(-0.101000)$

则 $[x]_{补}=11,101;00.110100$.

$[y]_{补}=11,100;11.011000$.

① 对阶 $[\Delta x-\Delta y]_{补}=00.001$

则 $[y]_{补}=11,101;11.101100$

则 $[-y]_{补}=11,101;00.010100$.

② $[S_{x\oplus y}]_{补}$ 为

$$\begin{array}{r} 00.110100 \\ +\ 00.010100 \\ \hline 01.001000 \end{array}$$

即 $[x-y]_{补}=11,101;01.001000$.

③ 规格化. $[x-y]_{补}=11,110;00.100100$